Multiple Docker containers are able to use just a single instance of the Linux kernel, and for this reason multiple containers are far more performant than multiple VMs. The isolation of containers from each other and the host operating system (via a Linux kernel feature called *namespacing*, a concept that also exists in PHP and we'll be learning about in great detail later) means that the container's view of the operating environment is restricted. Containers have their own process trees, users, filesystems, and network environment. Container access to host hardware devices (such as disk and network) is controlled by the docker server process `dockerd` (also known as the docker *daemon*, daemon processes in Unix and Linux operate in the background performing system level tasks). `dockerd` deals with grunt work of managing container life-cycles, mapping host filesystems into the container, creating virtual network interfaces so the container can access network services, and so on. Interaction with `dockerd` is usually through the `docker` CLI binary, or the Docker Desktop GUI.

CLI Installation

As with the PHP installations we saw in the previous section, your OS package manger should be capable of installing docker for you:

Debian/Apt

```
$ sudo apt-get update
$ sudo apt-get install -y docker.io docker-compose
$ sudo usermod -aG docker USER
```

Interacting with Docker on Fedora requires superuser privileges, the last command above adds `USER` to the `docker` group (change `USER` to whatever your user ID is) which gets around having to `sudo` every `docker` command. You might need to logout and back in again for it to take effect. See https://docs.docker.com/engine/install/debian/ for more details, or if you have any problems.

Fedora/DNF

```
$ sudo dnf -y install dnf-plugins-core
$ sudo dnf-3 config-manager --add-repo https://download.docker.com/linux/
fedora/docker-ce.repo
```